

US Equity Market-Neutral

Gradient-boosted trees on 147 firm characteristics, traded through a sector-, size- and beta-residualised long/short book

INFORMATION RATIO

+0.88

vs 3M T-bill + 4%, gross

IR AFTER 20BPS

+0.71

trading costs applied

BETA VS S&P 500

-0.159

se=0.120

MAX DRAWDOWN

-6.53%

monthly marks

Evaluation window **2021-01 to 2026-08 (68 months)** · 200 names · 200% gross · net +12.5%

The alpha changed legs mid-period. 2021-22 came almost entirely from the short book during the speculative unwind; since 2023 it has come from the long book. We do not claim the first is repeatable.

The strategy

Rank, then remove everything we are not trying to bet on

1 Screen

Price $\geq$ \$5, no nano or micro caps. Without it the short book fills with penny stocks: median cap \$50m, and -48% in January 2021 alone.

2 Residualise

The forecast is regressed on sector, size and beta each month; only the residual is traded. Neutrality is enforced on the signal, not patched onto the weights.

3 Smooth

Scores averaged over three months. Turnover falls from 56% to 24.6%; break-even cost rises to 93bps.

4 Size

Inverse-volatility weights inside each leg, capped near 2%. The information ratio is a ratio - the denominator counts.

5 Match beta

Leg notionals set from each leg's own realised sensitivity over 24 completed months, not from per-stock estimates, which ran 0.30 too high on the long side.

Largest average long positions

Ticker	Company	Avg wt
INCY	INCYTE CORP	1.13%
USNA	USANA HEALTH SCIENCES INC	0.78%
YELP	YELP INC	0.76%
NNI	NELNET INC	0.74%
GOOGL	ALPHABET INC	0.72%
HSTM	HEALTHSTREAM INC	0.69%
TNET	TRINET GROUP INC	0.67%
CORT	CORCEPT THERAPEUTICS INC	0.66%

Largest average short positions

Ticker	Company	Avg wt
FSLR	FIRST SOLAR INC	-0.73%
DAL	DELTA AIR LINES INC	-0.72%
LVS	LAS VEGAS SANDS CORP	-0.68%
UAL	UNITED AIRLINES HOLDINGS I	-0.60%
SWCH	SWITCH INC	-0.58%
VIRT	VIRTU FINANCIAL INC	-0.58%
PCG	P G & E CORP	-0.57%
AAL	AMERICAN AIRLINES GROUP IN	-0.55%

Data and method

147 supplied characteristics; the 8-K corpus measured and reported

Forecast

- Gradient-boosted trees, refit annually on an expanding window with a rolling two-year validation block, split by target month.
- Missing values are NOT imputed for the trees. Missingness is informative - stocks missing the most characteristics have a median cap of \$282m against \$1,687m.
- The target is demeaned within each month. Fitting raw returns stopped after ONE boosting round: training averaged +0.27%/month against +2.52% in validation.
- Model choice is made inside each fold on that fold's validation block only.

Out-of-sample R-squared, benchmarked against zero

Model	OOS R2	Monthly IC
Gradient-boosted trees	+0.3956%	+0.1354
Ridge	+0.0278%	+0.1193
OLS	-0.0068%	+0.0806
Lasso / Elastic Net	negative	+0.08

The rules note 1-2% is typical even for neural networks. A large positive number would mean a leak, not skill.

The 8-K corpus: measured, not assumed

- 27 single-signal tests on item codes. None survives a monthly cross-sectional regression with controls.
- Joined into the model: validation rank correlation +0.1087 without, +0.1082 with. No gain.
- The model's own usage agrees - the ten filing columns take 1.71% of its attention against the 6.37% an average feature would.
- One exception: item 5.02 is null in aggregate (+0.046%/mo, $t=+0.5$) but splits into abrupt departures at -0.640%/mo ($t=-1.9$) and routine appointments at +0.121%.

The trap worth knowing

Item 5.02's own heading contains every keyword worth searching for. Measured raw, "appoint" appears in 99.2% of these filings and the sub-item markers in 88-100%. Strip the heading and they fall to 54.5% and below. Any classifier trained on the unstripped text is learning the title.

Returns

2021-01 to 2026-08, 68 months, gross of trading costs unless stated

ANNUALISED (CAGR)

+19.65%

arithmetic +18.90%

CUMULATIVE

+176.42%

hurdle +50.87%

HIT RATE

61.8%

months beating the hurdle

MEAN MONTH

+1.575%

best +10.90% / worst -6.53%

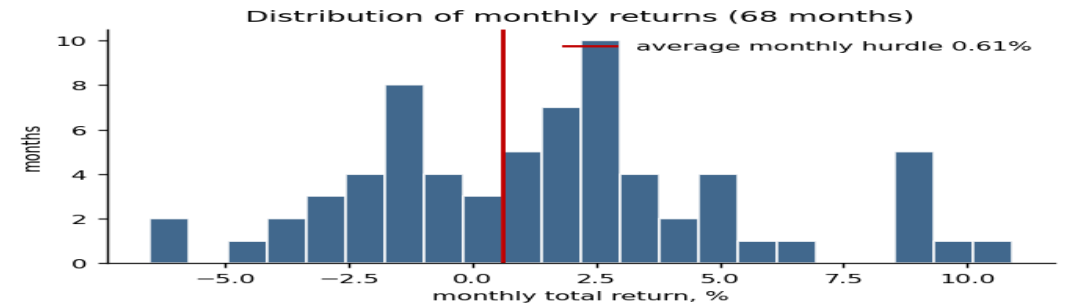
Calendar years

Year	Strategy	Hurdle	S&P 500
2021	+12.95%	+4.12%	+26.89%
2022	+38.51%	+6.19%	-19.44%
2023	+13.07%	+9.45%	+24.23%
2024	+8.89%	+9.34%	+23.31%
2025	+19.60%	+8.37%	+16.39%
2026	+19.99%	+5.21%	+12.28%

Where the return came from

Leg, measured against the eligible universe	2021-22	2023-26
Long leg excess	-3.07%	+8.81%
Long leg t-statistic	-0.5	+2.4
Short leg excess (negative is good)	-29.35%	-1.61%
Short leg t-statistic	-2.5	-0.2

Raw contribution conflates a rising market with poor selection. Measured against the universe we could actually trade, the legs take turns: the short book carried 2021-22, the long book carries 2023-26.



Risk-adjusted performance

The information ratio is the headline; beta is the entry requirement

INFORMATION RATIO

+0.88

vs cash + 4%, annualised

SHARPE

+1.19

over cash, annualised

ALPHA VS S&P 500

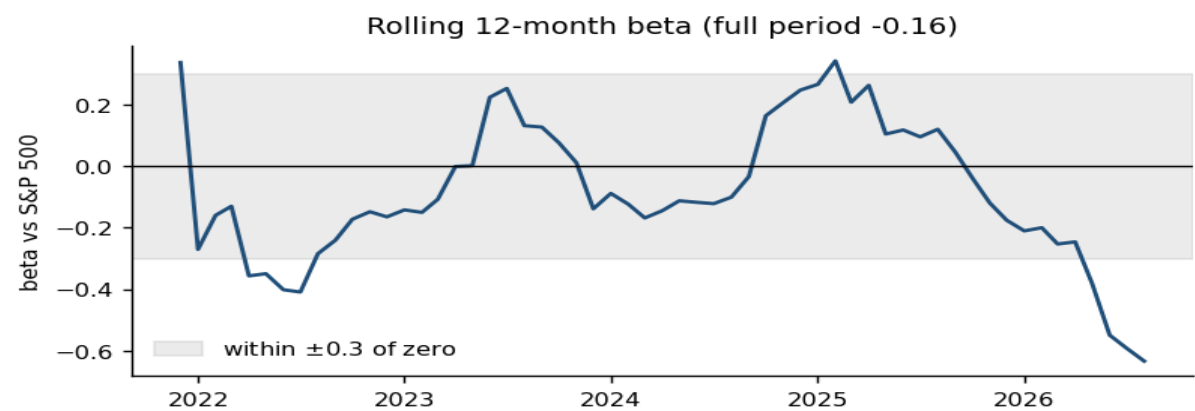
+17.29%

t=+3.22

BETA

-0.159

se 0.120



Outside +/-0.3: 10 of 57 windows, which is 3 distinct episodes once the overlap between 12-month windows is collapsed - 2021-01..2022-07, 2024-03..2025-02, 2025-06..2026-08.

None of them is distinguishable from zero (largest |t| 1.9). A 12-month window measures beta 2.0x looser than the full period, so a book with our full-period beta would leave the band in 32% of windows on sampling noise alone. We leave it in 18%.

Cost sensitivity

Cost per trade	Information ratio	Annualised
0 bps	+0.88	+19.65%
10 bps	+0.79	+18.29%
20 bps	+0.71	+16.93%
30 bps	+0.62	+15.60%
50 bps	+0.44	+12.96%

Break-even cost: 100 bps per trade

Borrow fees are not modelled. At 25bps trading plus 1%/yr borrow and a haircut on the collateral rate, the information ratio is roughly 0.5 - the number we would rather defend.

Exposure and implementation

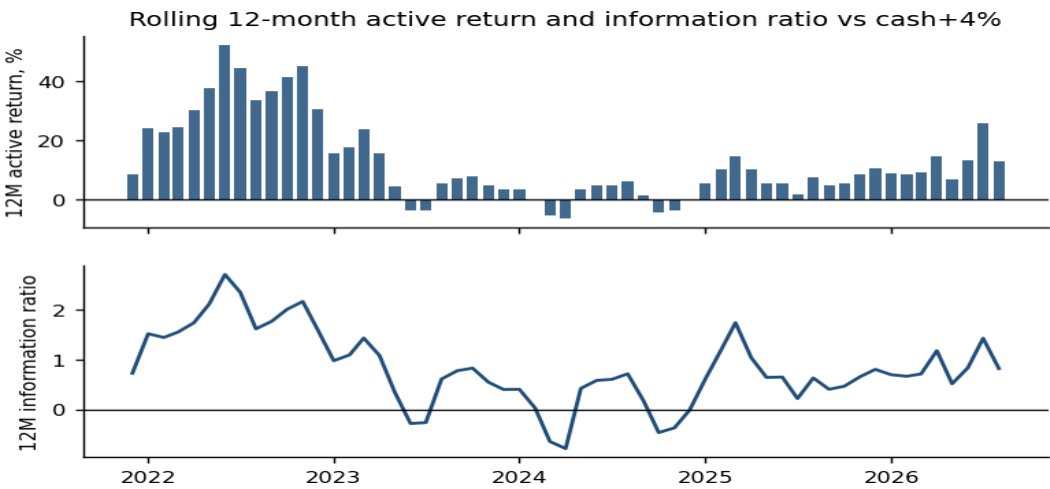
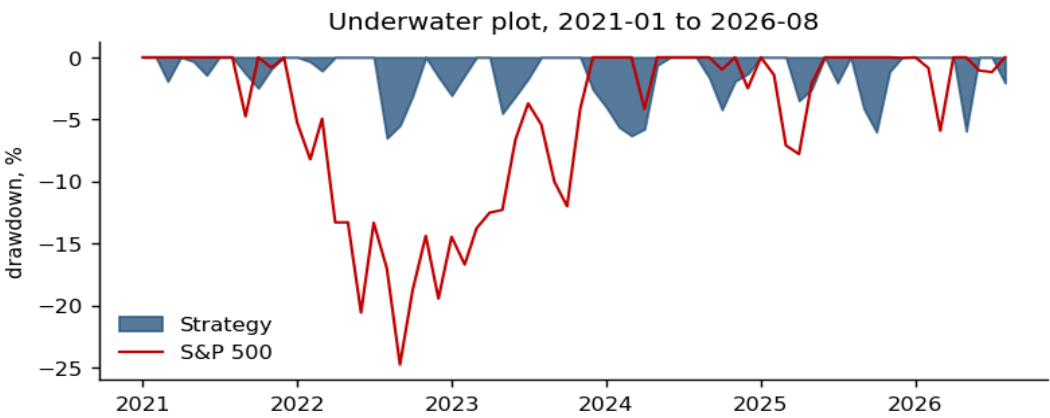
All four trading limits hold in every one of the 68 months

Measure	Value	Range / note
Holdings	200	100 long / 100 short
Gross exposure	200.0%	range 200.0%-200.0%
Net exposure	+12.5%	range -24.0% to +30.0%
Largest single position	2.24%	average 1.000%
Top 10 names	9.7%	share of gross
Monthly turnover	24.6%	range 17%-41%
Notional traded	99% of capital	per month

Short-book borrowability

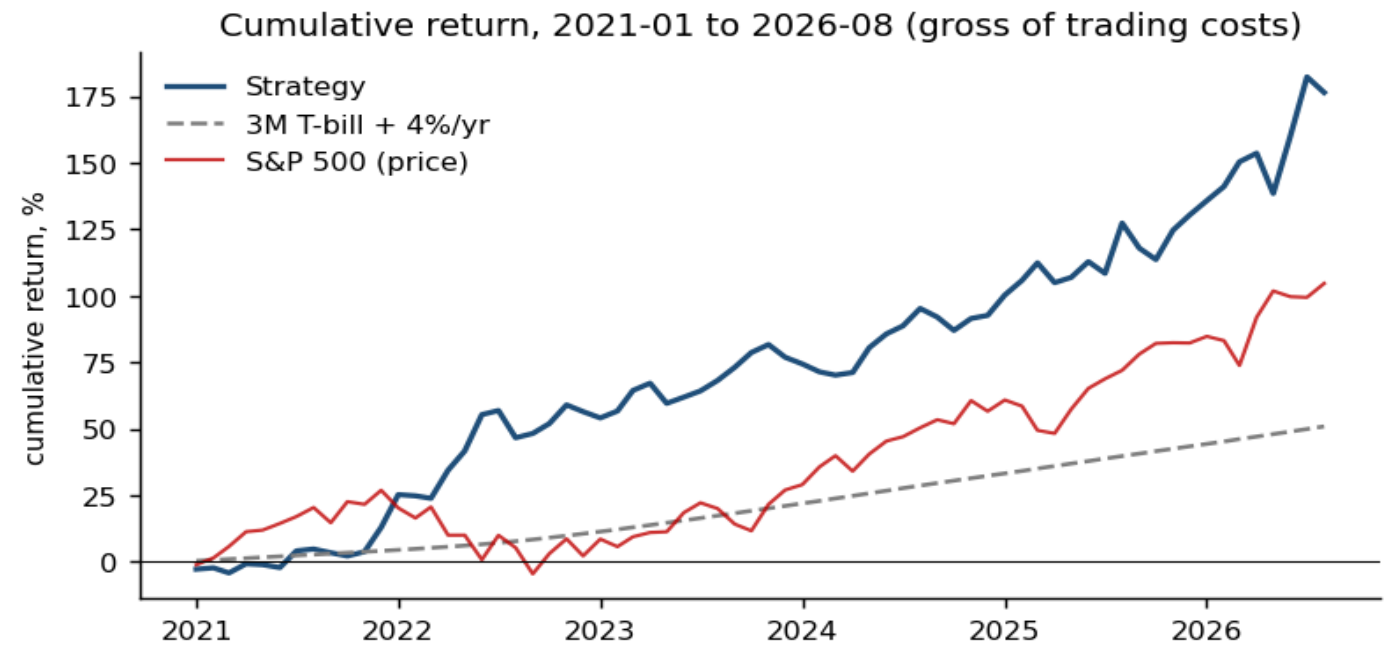
	Short leg	Long leg
Median market cap	\$3714m	\$2577m
Median daily dollar volume	\$1150.3m	\$559.3m
Median price	\$28.85	-
Small cap or below	48.7%	nano/micro screened out

The short book is more liquid than the long book. The unscreened version had a \$50m median cap and a \$2.63 median price.

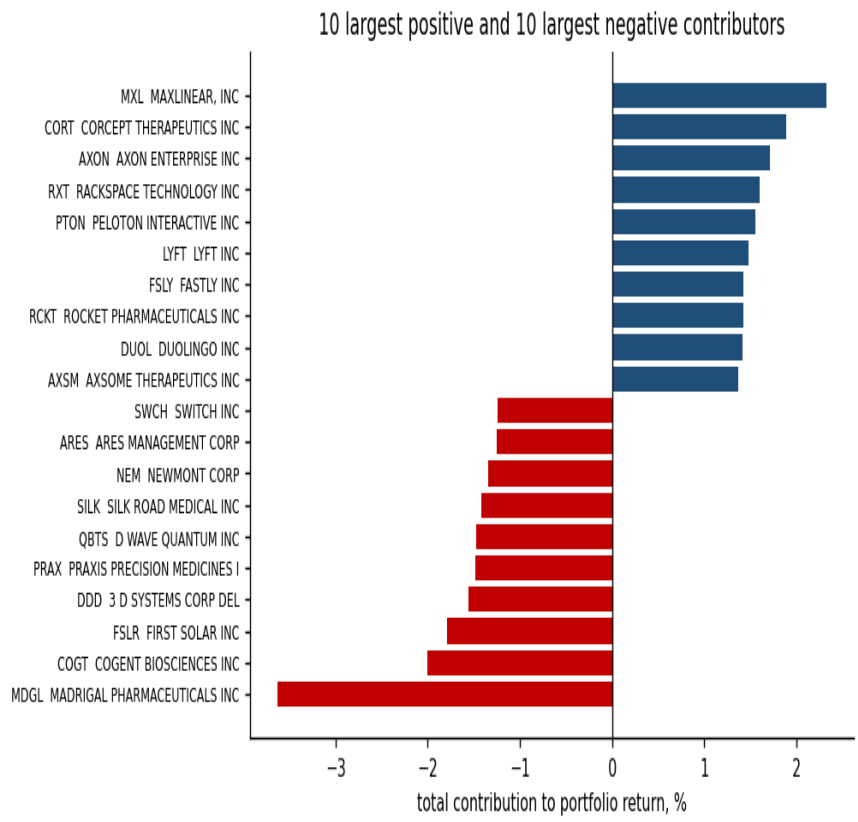


Cumulative performance

Strategy against the cash-plus-4% hurdle, with the S&P 500 for context



Beating the market is not the objective this year. The S&P 500 is shown because the rules ask for context and because the regression of our excess returns on the market's is how neutrality is verified. The number that decides the strategy is the distance from the grey hurdle line.



What we would tell an investment committee

What worked

- Portfolio construction moved the book from beta - 0.537 to -0.159 and cut drawdown from -30% to -6.53%.
- Alpha survives market, size and value: +13.78%/yr, $t=+2.47$, R^2 0.15. It is not a repackaged factor.
- The long leg has genuine selection skill since 2023: +8.81% over the eligible universe, $t=+2.4$ across 44 months.

What did not

- The short leg's edge was one regime. In the pre-test validation window it pointed the wrong way (+5.42% vs universe); since 2023 it is statistically zero.
- 8-K signals add nothing measurable to the forecast, across three independent tests.
- Removing the 5 best months of 68 takes the information ratio from +0.88 to +0.36.

What we would say about the future

- The second half of the window runs at +0.62 against +1.12 in the first. The later figure is the more honest forward expectation; 2021-22 dominates the full-period number.
- The short book's job is neutrality, not return. We would size it as a hedge, not as a second source of alpha.
- We evaluated the test period 13 times. Nine followed a diagnosed defect; three were genuine trials, one of which we removed after it failed.
- The improvement we would make next is sector neutrality: 90% of capital sits in sector bets nobody chose. We tested it four ways and report it as untested rather than rejected - the appendix explains why our sample cannot resolve an effect that size.

Every one of these was measured, not asserted. The appendix carries the regressions, the research log and the compliance output.

Where the alpha came from, and why it moved

Each leg measured against the universe it was picked from, which removes market direction

Period	Universe	Long vs universe	Short vs universe
2019-20 (pre-test validation)	+25.03%	+0.85% (t +0.1)	+5.42% (t +1.1)
2021-22	-1.57%	-3.07% (t -0.5)	-29.35% (t -2.5)
2023-26	+12.05%	+8.81% (t +2.4)	-1.61% (t -0.2)

- For the short leg a negative number is good: the names we shorted underperformed.
- The only window with short-side skill is 2021-22, the speculative unwind. In the pre-test validation window the short leg was the wrong way round.
- Splitting the short book by market cap, price and volatility, EVERY bucket flips positive to negative. No screen repairs it: dropping the extreme-volatility names would have cost 18.8pp of the 2021-22 gain to recover 3.7pp of the later loss.
- The 2021-22 short alpha survives sector adjustment almost intact (-26.90% vs -25.62%), so it was stock selection inside sectors, not a bet against technology.

Applying the same standard to both legs

The long leg also shows skill in only one regime. If we discount the short result as regime-specific, the same scepticism applies to the long one. The framing we prefer is that the legs take turns: shorts pay when speculative names unwind, longs pay when quality leads.

Text data: three tests, one null, one exception

This year's theme, measured rather than assumed

Test	What we found
27 single-signal tests on item codes	None clears a monthly regression with controls
Joined into the forecast	Validation IC +0.1087 without, +0.1082 with
The model's own feature importance	10 filing columns take 1.71% against 6.37% expected
Item 5.02 split by triage	Aggregate +0.046% (t +0.5) -> abrupt -0.640% (t -1.9)

- The naive t-statistic was inflated in all of these. Stocks within a month move together, so 7,670 stock-months are not 7,670 independent observations. Three signals that looked alive on the naive test died on the monthly one.
- Coverage is the binding constraint: only 50.7% of evaluation-window stock-months have any filing, and the abrupt-departure flag is set on 1.26%. A tree splits on what separates many rows; a rare flag is structurally ignored however accurate it is.
- The triage uses no language model. Rules were read off the documents; --sample prints both sides for inspection and sorted 8 of 8 correctly on the sample shown.

Why the aggregate hid it

70,582 officer-change filings carry no signal together. Separated, 1,632 abrupt departures move next-month returns by -0.640% while 14,746 routine appointments move them +0.121%. An aggregate null is not the same as no signal - it can be a diluted one.

t = -1.9 still does not clear the bar we set for ourselves after this many tests, and we say so.

Robustness

Six tests, run before the deck was written

Subperiod	Information ratio	vs hurdle
2021 rebound	+0.72	12 months, +8.8% vs hurdle
2022 drawdown	+1.58	12 months, +32.3% vs hurdle
2023-25 concentration	+0.46	36 months, +17.6% vs hurdle
2026 so far	+1.23	8 months, +14.8% vs hurdle
first half	+1.12	34 months
second half	+0.62	34 months

Stress test	Information ratio
all months	+0.88
minus the 1 best months	+0.79
minus the 3 best months	+0.59
minus the 5 best months	+0.36
minus the 5 best stocks	+0.76
minus the 10 best stocks	+0.67
minus the 20 best stocks	+0.50

Factor attribution

Model	Annualised alpha	t and R2
market only	+18.30%	t=+3.37, R2 0.05
market + size + value	+13.78%	t=+2.47, R2 0.15

- Loadings on size and value are insignificant, and the three-factor regression leaves most of the variation unexplained (t=+2.47, R2 0.15).
- Delisting marks: positions with no realised return are marked at zero in the base case (50 of 13,600, 0.64% by weight). At Shumway's -30% convention the net information ratio is +0.54.
- Daily marks cover 82.3% of the book by weight and 64.4% at worst. Names that cannot be resolved were acquired or renamed, so daily figures are biased optimistic.
- Maximum drawdown on daily marks is -11.93% against -6.53% on monthly marks. Rolling 12-month beta averages -0.08 (range -0.63..+0.34, outside +/-0.3 in 18% of windows).

Research log and multiple testing

The question we expect to be asked, answered before it is

- We evaluated the test period 13 times. Nine of those followed a diagnosed defect and had a reason independent of the result: penny stocks in the short book (-48% in January 2021), beta at -0.537, the wrong beta estimator, the tree target not demeaned (one boosting round per fold), leg beta measured rather than estimated, unstable validation tuning, and a cross-fold leak in model selection.
- Three were genuine trials: the tree model, turnover control, and the 8-K feature set. We removed the third after it failed its pre-declared test.
- One correction is worth naming. `choose_blend` pooled all six folds' validation blocks before picking one model for all six years, so the 2021 forecast was chosen partly on 2024-25 outcomes. Each fold's own split was clean and our checks passed - the leak lived between folds, where a per-fold check cannot see it. Selection now happens inside each fold, and a new check fails on the old format.
- What we take from this: with 68 months and this many looks, a t-statistic near 2 is not evidence. What makes a result worth keeping is effect size, a reason to expect it in advance, and survival across regimes - not the t.

Reproducibility

MAIN.py runs the whole chain from the supplied parquet files to the compliance gate and the exhibits on these slides, and exits non-zero if any rule is broken. 22 automatic checks cover look-ahead (predictor list, split keying, preprocessing fit window, cross-fold selection) and the trading criteria (100-500 names, gross $\leq 200\%$, net inside $\pm 50\%$, both legs, coverage, tradability). A separate test feeds the checker 21 deliberately broken inputs and asserts each is caught; all 21 fire. The traded book records 0 failures across all 68 months.

A proposed improvement we tested and did not adopt

Sector neutrality: what the experiments said, and why they cannot settle it

The book carries a sector bet nobody chose

- Residualising the score on sector dummies removes each sector's mean, not its skew. The traded book averages 8 technology longs against 28 shorts, and 89.7% of capital sits in net sector positions.
- Shrinking those nets toward zero and restoring gross traces the whole curve:

Neutralisation	Net sector	Spread	Vol	IR
none (traded book)	89.7%	+15.62%	13.15%	+0.88
a quarter removed	73.2%	+16.16%	13.04%	+0.93
half removed	53.6%	+16.87%	13.14%	+0.98
three quarters	29.9%	+17.85%	13.60%	+1.02
fully neutral	0.0%	+19.28%	14.68%	+1.04

Volatility is flat until sector exposure falls below about half of capital, then rises. Variant S sits at 38.2%, where the curve predicts its volatility to within 0.11pp - so its flat vol is not a puzzle. Its return, though, beats the curve by 4.2pp: only about a third of variant S's gain is the sector exposure it removes.

What this evidence can decide, and what it cannot

Exact. Sector net exposure falls from 89.7% of capital to 38.2% of capital. That is arithmetic on the weights and carries no sampling error at all.

Estimated, and not established. Market sensitivity is the mandate metric, and it does not hold up: 28 of 57 rolling windows improve (49% -- a coin flip is 28), full-period |beta| moves -0.034 — 0.28 of one standard error — and the headline last window, -0.63 to -0.21, is 1.3 standard errors -- not significant.

Not resolvable. Validation cannot detect an IR gap below 1.12, the test period below 0.66; every effect here is between +0.2 and +0.4, and our own conclusions reversed three times. The book is unchanged: adopting would be a judgement about mandate exposure, not a statistical finding, and we would say so.

Four rounds of testing, three reversals

- Picking inside sectors loses on the clean 2019-20 window and gains nothing on the test period - until the quota rule, smoothing and buffer are varied, after which it leads in 7 of 18 configurations (binomial $p = 0.48$). Under a true null the best of 18 reaches ours essentially always.
- A no-sector rule that evicts the weakest names beat the sector cap at one smoothing setting and lost at the other two, net of costs. We withdrew that conclusion too.

Net of 20bps	Validation	Test period
Traded book	-0.66	+0.67
Sector cap 5	-1.05	+0.91
Evict 10/month	-1.32	+0.97

Every arm beats the traded book on the test period and loses to it on the only window the model never saw. Exploratory arms were run on an earlier build (baseline +0.85); the dose curve and variant S are on the submitted book.